



The national standards for cancer survivorship care in action: caregiver experiences across stage, phase, and geography

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Abstract

Purpose Caregivers play a central role in supporting cancer survivors across the care continuum, often at detriment to their own well-being. Recognizing caregivers' role in survivorship, the National Standards for Cancer Survivorship Care state the health system should assess "caregivers' experiences and unmet needs." This study sought to describe the extent to which this standard is being met from the perspective of caregivers.

Methods This cross-sectional, convergent mixed-methods study leveraged survey and semi-structured interview data from caregivers of survivors in active and post-treatment survivorship care at a southeastern U.S. cancer center. Surveys queried caregivers' concerns and their experience being asked about and communicating various concerns; interviews further characterized these experiences, including barriers and facilitators to accessing services. Survey data were analyzed descriptively (and by various subgroups), and interview data were examined using directed content analysis.

Results For the 77 caregivers surveyed (64.94% female; 96.10% non-Hispanic White; M average caregiving/week = 36 h), emotional concerns were most reported (66%), yet $\geq 50\%$ of caregivers reported never being asked about these issues by the care team. Rural caregivers were less likely than non-rural caregivers to report the care team had inquired about their social and medical caregiving concerns. Interview data highlighted limited assessment of caregiver needs, logistical and communication barriers, and desire for proactive, tailored supportive services.

Conclusions Practice gaps in assessing and addressing caregivers' emotional, social, and physical health needs must be overcome to meet the caregiver standard, particularly if assessment is to be comprehensive in scope to mirror survivor assessment standards.

Implications for Cancer Survivors Cancer care should implement routine, comprehensive structured assessment and referral pathways to meet caregiver needs.

Keywords Cancer survivors · Caregivers · Standards

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Introduction

By 2040, the United States (U.S.) healthcare system will need to meet the needs of 26 million cancer survivors, an estimated 55% of whom rely on unpaid family and friends (i.e., caregivers) to provide critical medical, instrumental, and psychosocial support [1]. Meeting survivors' needs requires meeting caregivers' needs [2], as they are an essential part of the healthcare team throughout the cancer survivorship trajectory.

Across the cancer survivorship trajectory, caregivers play an indispensable role in supporting survivors' physical, emotional, and practical needs. During active treatment [3], caregivers attend appointments, manage medication schedules, provide

emotional support, and help with daily tasks. Therapeutic advances have only increased the demands of caregiving, as caregivers have greater responsibility for monitoring treatment side effects at home, adherence to oral agents, and sustaining this role throughout longer disease trajectories [4–6]. In the post-treatment phase, many caregivers assist and support survivors with long-term symptom management, surveillance for recurrence, and adjustment to life after cancer [7]. As evidenced by high rates of psychological distress and financial toxicity [8], caregiving is often time-consuming, emotionally taxing, and disruptive to caregivers' own health, employment, and personal goals [9]. An estimated 42% of caregivers experience depression, 46% anxiety, and 25% financial strain [10–12].

Despite their critical role and high need for services to address the burdens of caregiving, caregivers' needs remain to be systematically identified and addressed in cancer care [13]. In 2024, the National Cancer Institute introduced the National Standards for Cancer Survivorship Care to promote high-quality, coordinated, and person-centered care across the survivorship continuum [14]. The national standards outline essential components, including care planning, service provision, and system-level infrastructure, and recognize caregiver outcomes as an important part of survivorship care, requiring that the health system has a “process to collect data on caregivers' experiences and unmet needs.” Despite the promise of these standards, there is little evidence on how they are operationalized in practice to meet caregiver needs. Understanding how survivorship care is implemented and the extent to which caregivers are meaningfully included is essential to improve outcomes for survivors and caregivers alike.

The current study aims to evaluate the state of caregiver-specific cancer survivorship care at an NCI-Designated Comprehensive Cancer Center, understand caregivers' barriers to care, and explore differences in caregiver experiences by treatment phase (active vs. post-treatment) [3, 15], cancer stage (early vs. advanced) [16, 17], and geographic residence (rural vs. non-rural) [16, 18–20]. Findings from the study will identify priorities for meeting the new NCI National Standards for Cancer Survivorship Care among caregivers.

Methods

Study design

This cross-sectional, convergent mixed-methods study utilized quantitative survey data and qualitative interview data to describe caregiver experiences of survivorship care at an NCI-Designated Comprehensive Cancer Center in the south-eastern U.S. Caregivers were recruited as part of a larger protocol in which survivors were also approached to complete a survey and interview (see Webster et al. (this issue)).

Participants

Participants were caregivers of survivors receiving care from one of four outpatient clinics (breast, gynecologic, head and neck, and multidisciplinary clinic) for solid tumors; survivors were at least 3 months into active treatment or were in clinic for post-treatment survivorship care. Caregivers were eligible if their care recipient agreed to their recruitment into the study, they self-identified as a friend or family member who was the primary source of cancer-related care and support, were at least 18 years old, and were able to understand and complete a survey and interview questions in English.

Procedures

Caregivers were identified at the cancer care appointments of their care recipient. We purposively sampled to ensure that caregivers represented relatively even distributions regarding survivors' EHR-documented treatment phase (active vs. post-treatment) and disease stage (early vs. advanced). To recruit a caregiver, study staff first briefly described the caregiver study to the survivor and obtained permission verbally from the survivor to contact the person they identified as a primary support person (i.e., caregiver). In most cases, the caregiver was present during the clinic visit, which allowed study staff to screen and enroll the caregiver during the same visit. When the caregiver was not present, the survivor provided the name and contact preferences for the caregiver. Study staff subsequently contacted the caregiver via their preferred communication method (phone, email, text) to describe the study. Interested caregivers were screened for eligibility and gave verbal consent. Participating caregivers were asked to complete a 15-min survey either in clinic (on paper or a tablet) or via an online form after the visit. Participants received \$50 for completing the survey. Survey data were collected between February and June 2025. Caregivers who indicated interest in completing an interview during the verbal consent process were purposively sampled by clinic type, sex, rural status, and care recipient's cancer treatment status (active vs. post-treatment). Some interview participants completed both the survey and interview, whereas others completed the interview only. Caregivers were contacted by phone to complete semi-structured interviews between April and July 2025. Interviews were audio-recorded using HIPAA-compliant Zoom and transcribed verbatim, with transcripts cleaned using Rev software. Participants received \$50 for the interview. Study procedures were approved by the university's Institutional Review Board (IRB #98262).

Measures

Survey

Sample characteristics Participants were asked standard sociodemographic questions and identified their relationship to their cancer care recipient. Rural residence was determined by comparing the county corresponding to caregivers' self-reported ZIP code to the 2023 USDA Rural-Urban Continuum Codes, which assigns U.S. counties a code from 1 to 9, with codes 4–9 defined as rural [21]. To assess participants' role as a caregiver, they were asked: "Do you help the person for whom you are providing care with any of the following activities? Check all that apply." This statement was adapted from the HINTS 5 Cycle 2 Caregiver Module [22], with options such as "Performing medical/nursing tasks," "Communicating with health care professionals," and "Providing transportation." Caregivers were also asked to report on their quality of life using the Caregiver Roles and Responsibilities Scale (CRRS), which consists of five subscales and items that have five response options: "Not at all," "A little bit," "Somewhat," "Quite a bit," or "Very much" [23]. The Support and Impact subscale (six items, possible score range 0–24; Cronbach's $\alpha=0.84$) assesses emotional and practical support that caregivers receive from others. The Lifestyle subscale (11 items, possible score range 0–44; Cronbach's $\alpha=0.88$) assesses the impact of caregiving on daily routines, social life, leisure, and health. The Emotional Health and Wellbeing subscale (nine items, possible score range 0–36; Cronbach's $\alpha=0.91$) assesses psychological strain related to caregiving. The Self-Care subscale (six items, possible score range 0–24; Cronbach's $\alpha=0.78$) assesses attendance to their own health and personal needs. The Financial Wellbeing subscale (six items, possible score range 0–24; Cronbach's $\alpha=0.83$) assesses the financial effects of caregiving. Subscale scores were calculated by summing items, and a total score (possible score range 0–152) was calculated as the sum of the five subscale scores. Higher total scores indicate better overall quality of life, and higher subscale scores reflect greater well-being or less burden within each domain.

Experiences related to national standard Because the National Standards do not specify which experiences and unmet needs health systems should assess [14] and in an effort to aid future comparison with any reports of *survivor* assessment practices, we included domains that mirrored applicable survivor domains in the standards, supplemented with caregiving-specific domains from prior research (Table 1) [24–27]. Caregivers were asked how often (during the survivor's clinic visits) the cancer care team asked about their concerns (emotional, physical, and social) and

challenges in providing medical and nonmedical care to survivors. For example, they were asked: "Since the person for whom you are providing care was diagnosed, how often has the cancer treatment team asked (either in conversation or a form) about whether you are experiencing emotional strain as a caregiver?" Response options ranged from "has asked me at every visit" to "has not asked me." These items were adapted from the Patient Centered Survivorship Index [28]. Participants reported how often they felt that the care team listened to their emotional, physical, and social concerns from "not at all" to "completely." They were asked whether they had experienced concerns in each domain, and whether they received any advice or referrals for managing their concerns as yes/no items.

Interview

A standardized interview guide (Supplemental File 1) was developed based on the NCI's National Standards for Cancer Survivorship Care [14]. Questions centered on caregivers' personal experiences, such as "During [your family member/friend]'s treatment at [cancer center], how – if at all – has the care team checked in with you about your own needs as a caregiver?" Some questions assessed caregivers' involvement in communication with providers, such as "How well did you feel that [your family member/friend]'s care team engaged you in the planning process by discussing shared goals of care and coordination with other health services?" Sample follow-up probes could be tailored to participants' responses.

Data analysis

EHR and survey data

We used descriptive statistics to summarize the demographics, disease-related characteristics, and quality of life of the study sample. Primary outcomes included the proportion of caregivers who reported experiencing each domain of concern (e.g., emotional, physical, social), the proportion who indicated that they had been asked about these concerns, and the proportion who communicated their concerns to the cancer treatment team. Communication percentages were calculated among those who reported experiencing the corresponding concern. Group differences in these outcomes were examined by cancer stage (early vs. late), treatment phase (active vs. post-treatment), and geographic residence (rural vs. non-rural). Chi-square tests were used to assess associations between categorical variables, and Fisher's exact tests were applied when more than 20% of expected frequencies were less than five. All analyses were conducted using IBM SPSS Statistics, version 38, with statistical significance set at $p < 0.05$ (two-tailed).

Table 1 Comparison of survivor standards defined in the national standards and domains assessed in the caregiver survey

National standards area	Survivors	Caregivers
Health system evaluation/assessment	-	The organization has a process to collect data on... caregivers' experiences and unmet needs
health system processes	<i>National standards</i>	<i>Assessed in study's caregiver assessment</i>
Cancer survivors are...	Provided with access and referral to a survivorship program that addresses the needs of cancer survivors either on-site, through telehealth, or by referral Assessed at multiple points in their follow-up care for physical effects during and following cancer treatment, including monitoring for late effects and chronic conditions, and provided with treatment and/or referrals Provided with access to appropriate specialty care services to manage potential late effects (e.g., cardiovascular issues) either on-site, through telehealth, or by referral Assessed at multiple points in their follow-up care for emotional and psychological effects of cancer and its treatment and provided with treatment and/or referrals Assessed for practical and social effects of cancer and its treatment (e.g., social risks, health-related social needs, education and employment/return to work or school) and provided with resources and/or referrals Assessed for their risk of recurrence or new cancers, including family history and genetic testing, and provided with recommendations and referrals regarding surveillance for recurrence or new cancers Assessed for lifestyle behaviors and provided with recommended strategies for management and appropriate referrals or education as needed (e.g., smoking cessation, diet/nutrition counseling, promoting physical activity) Provided with access and referrals to appropriate supportive health services (e.g., nutrition, occupational and physical therapy, rehabilitation, sexual health, fertility services, dental and podiatry services) Assessed for financial hardship/toxicity and concerns regarding insurance coverage, and provided with resources and support as needed	Assessed for physical effects: • How often asked, how carefully listened to • Have experienced physical effects • If experienced, communicated with treatment team ◦ Treatment team has advised or helped them manage ◦ Referred to specialists or services, if needed ◦ If referred, have had difficulty accessing referral Assessed for negative impacts to social well-being: • How often asked, how carefully listened to • Have experienced negative impacts to social well-being • If experienced, communicated with treatment team ◦ Treatment team has advised or helped them manage ◦ Referred to specialists or services, if needed ◦ If referred, have had difficulty accessing referral Assessed for physical effects: • How often asked, how carefully listened to • Have experienced emotional strain • If experienced, communicated with treatment team ◦ Treatment team has advised or helped them manage ◦ Referred to specialists or services, if needed ◦ If referred, have had difficulty accessing referral Assessed for supportive services utilization: • Asked if interested in/needed specific services (e.g., support groups, financial counseling, or navigation services) • If interested in/needed a service, have accessed this service • If accessed service, accessed at UK or at a non-UK facility

Table 1 (continued)

National standards area	Survivors	Caregivers
	Engaged in the care planning process including discussion of shared goals of care, advanced care planning, and coordination of care with providers and services (e.g., primary care provider, other health professionals, and community-based services) as needed	Assessed for challenges related to tasks for medical care and daily living: • How often asked, how carefully listened to • Have experienced challenges • If experienced, communicated with treatment team ◦ Treatment team has advised or helped them manage ◦ Referred to specialists or services, if needed ◦ If referred, have had difficulty accessing referral Assessed for challenges related to non-medical caregiving tasks (e.g., arranging transportation, managing finances): • How often asked, how carefully listened to • Have experienced challenges • If experienced, communicated with treatment team ◦ Treatment team has advised or helped them manage ◦ Referred to specialists or services, if needed ◦ If referred, have had difficulty accessing referral

Interview data

Directed content analysis [29] was used to analyze qualitative data. One coder (SBH) developed the first version of the codebook based on the national standards [14]. Two team members (JLB and MAW) reviewed the codebook and suggested some revision to the organization and operational definitions for four main parent codes (survivorship care, suggestions and recommendations, facilitators and barriers, and stories) and subsequent child and grandchild codes. SBH and MAW completed one round of consensus coding with three transcripts, followed by a second round with an additional three transcripts. Initial coding agreement was assessed at the coded passage level, with coders independently applying codes before comparing coding. Initial agreement exceeded 80%, and any remaining discrepancies were resolved through discussion until consensus was reached. One additional child code was added to the codebook during this process. The remaining transcripts were coded independently. NVivo version 15 (Lumivero, Denver, CO) was used for coding and analysis. Finally, MAW and JLB conducted thematic analysis to identify common themes within codes and illustrative quotes for each code.

Results

Sample characteristics

Caregivers who participated in the survey ($N=77$) ranged in age from 24 to 84 years ($M=55.16$, $SD=14.34$). The sample was predominantly White (96.10%) and non-Hispanic (98.70%). Most caregivers were female (64.94%) and rural residents (59.74%). Caregivers were most often survivors' spouses (54.55%) or adult children (22.08%). There was a good distribution of caregivers with respect to survivor treatment phase (61.04% active treatment) and stage (58.99% late stage). On average, caregivers reported 35.83 h of care per week ($SD=42.96$; range 0–168). On the Caregiver Roles and Responsibilities Scale, caregivers reported moderate overall quality of life ($M=123.91$, $SD=27.52$). Among subscales, caregivers reported the best quality of life with respect to self-care and emotional health/well-being, and worst with respect to support impact (Table 2). Eight caregivers completed the semi-structured interview, with two caregivers completing the interview only (demographic data not available). Mean age of interview participants was 52 years ($SD=14.0$); four were female, three were rural residents, and four were providing care to someone on active treatment.

Table 2 Sociodemographic characteristics of caregivers who completed surveys (*N* = 77)

Variable	<i>n</i> (%) / mean (SD)
Age	55.16 (14.34); median = 57.0; range 24–84 years
Race	
White	74 (96.10%)
Black	2 (2.60%)
Asian	0 (0.00%)
American Indian/Alaska Native	1 (1.30%)
Missing	0 (0.00%)
Hispanic ethnicity	1 (1.30%)
Female gender	50 (64.94%)
Rural residence	46 (59.74%)
Education	
Less than high school	4 (5.19%)
High school diploma or GED	23 (29.87%)
Technical or vocational school	2 (2.60%)
Some college	13 (16.88%)
College graduate	23 (29.87%)
Post-graduate	11 (14.29%)
Missing	1 (1.30%)
Annual household income	
< \$15,000	7 (9.09%)
\$15,000–49,999	19 (24.67%)
\$50,000–99,999	23 (29.87%)
≥ \$100,000	24 (31.17%)
Missing	4 (5.19%)
Current employment status	
Working full time	29 (37.66%)
Working part time	1 (1.30%)
Homemaker	4 (5.19%)
Retired	26 (33.77%)
Disabled	4 (5.19%)
Unemployed	11 (14.29%)
Missing	2 (2.60%)
Quality of life	
Total	123.91 (27.52)
Support impact	15.86 (5.38)/24
Emotional health/well-being	26.89 (8.27)/36
Financial	16.29 (6.12)/24
Lifestyle	29.82 (9.49)/44
Self-care	18.62 (4.37)/24
Relation to care recipient	
Spouse	42 (54.55%)
Partner, unmarried	3 (3.90%)
Parent	4 (5.19%)
Adult child	17 (22.08%)
Sibling	7 (9.09%)
Friend	2 (2.60%)
Other	2 (2.60%)

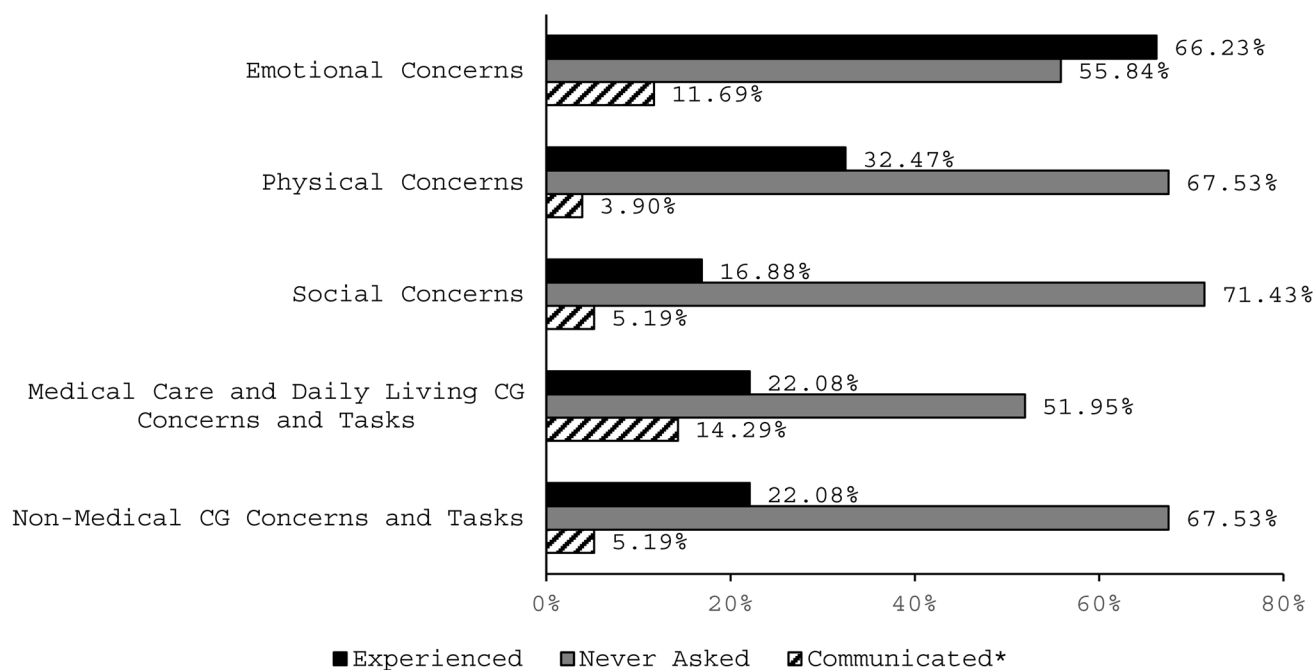
Table 2 (continued)

Variable	<i>n</i> (%) / mean (SD)
Survivor treatment status	
Active	47 (61.04%)
Post-treatment	30 (38.96%)
Survivor type of cancer	
Breast	17 (22.08%)
Gastrointestinal	17 (22.08%)
Genitourinary	5 (6.49%)
Gynecologic	0 (0.00%)
Head and neck	11 (14.29%)
Lung	17 (22.08%)
Skin	3 (3.90%)
Soft tissue and/or bone	4 (5.19%)
Unknown	1 (1.30%)
Survivor cancer stage	
Early (0–II)	26 (33.77%)
Late (III–IV)	45 (58.44%)
Unknown/unstaged	4 (5.20%)
Missing	2 (2.60%)
Survivor treatment type	
Surgery	10 (12.99%)
Radiation	12 (15.58%)
Chemotherapy	30 (38.96%)
Immunotherapy	24 (31.17%)
Other	7 (9.09%)
Current hours per week caregiving	35.83 (42.96) median = 20.0; range 0–168

Experience, assessment, and communication of caregiver concerns

The most common concerns caregivers reported were emotional (66.23%), followed by physical (32.47%); 22.08% reported concerns related to caregiving-specific medical (e.g., giving medications, communicating with the medical team) and non-medical caregiving tasks (e.g., arranging transportation to appointments, managing finances). Across all domains, at least half of caregivers reported that they had never been asked by the cancer care team about relevant concerns. Among caregivers who experienced concerns, 14.29% or fewer reported raising their concerns with the care team. Less than 6% of those who experienced physical concerns, social concerns, or non-medical caregiving tasks reported raising the concern with the care team (Fig. 1).

Comparisons by cancer stage, treatment phase, and geographic residence revealed few statistically significant differences in experiences or communication of concerns (Table 3), though many comparisons were notable. For example, 73.33% of caregivers of late-stage cancer survivors reported



*Communicated concerns were only asked about if the caregiver reported experiencing a concern, reducing the sample size.

Fig. 1 Experience, assessment, and communication of caregiver concerns^a

experiencing emotional concerns (vs. 53.85% of early stage); 26.67% of these caregivers reported medical caregiving-related concerns (vs. 11.54% of early stage). Similarly, caregivers of survivors in active treatment reported higher rates of experiencing physical concerns themselves (40.43% vs. 20.00% post-treatment) and somewhat higher emotional concerns (70.21% vs. 60.00% post-treatment). Rural caregivers reported lower rates of social concerns (13.04% vs. 23.33% among non-rural caregivers) and lower rates of medical caregiving-related concerns (19.57% vs. 29.99% among non-rural caregivers).

Statistically significant differences emerged for care team inquiry: caregivers from non-rural areas (vs. rural) were more likely to report that the care team asked at least half of the time about social (23.33% vs. 8.69%; $\chi^2 = 6.98$, $p = 0.02$) and medical care-related (29.99% vs. 19.57%; $\chi^2 = 7.59$, $p = 0.02$) concerns. No significant differences were observed in the communication of caregiving-related concerns across any groups.

In the interviews, caregivers described their experiences with and perspectives about the survivorship care process. Overall, few caregivers expressed awareness of or direct engagement with explicit survivorship care services such as meeting with a survivorship care nurse. Caregivers noted that information about supportive care services was often provided through printed resources at the beginning of treatment, which then necessitated proactive follow-up. Caregivers highlighted efforts of nurses and support staff to frequently check in about survivors' needs. Caregivers noted

that the care team often asked them directly if they had questions or concerns about the survivor's care (Table 4, section A); however, caregiver needs were not assessed directly, if at all (Table 4, section B). Despite this, caregivers felt connected with providers and were involved in the care planning process for their care recipient, noting that if concerns were identified, they were adequately addressed (Table 4, C–E).

Caregivers noted several salient barriers within the health system. These included challenges related to physically navigating multiple appointments within the hospital space and parking, coordinating care between multiple specialty providers, and poor communication from providers and feeling as if their concerns about survivors' care were not fully appreciated (Table 4, F). Caregivers noted several internal facilitators to survivorship care, including approachable providers, effective coordination of interdisciplinary care, and integration of supportive care services into treatment so that caregivers needed to seek out fewer resources (Table 4, G–H). External to the health system, caregivers noted that physical distance from the cancer center and transportation were barriers to accessing survivorship care (Table 4, I). Caregivers noted barriers due to financial limitations or insurance challenges, difficulty taking family leave, and issues coordinating care between the cancer center and local clinics (Table 4, J). External facilitators of survivorship care shared by caregivers included effective insurance coverage, charitable financial and logistical supports, and local access to certain healthcare services (Table 4, K).

Table 3 Experience, assessment, and communication of caregiver concerns by group

	Total N = 77	Cancer stage		χ^2 (df); p	Treatment phase		χ^2 (df); p		Geographic residence	
		Early n = 26	Late n = 45		Active n = 47	Post-treatment n = 30			Rural n = 46	Non-rural n = 30
Emotional experienced	51 (66.23%)	14 (53.85%)	33 (73.33%)	2.32 (1); 0.13	33 (70.21%)	18 (60.00%)	0.85 (1); 0.36	30 (65.22%)	20 (66.67%)	0.02 (1); 0.90
Emotional asked										
Never	43 (55.84%)	13 (50.00%)	26 (57.78%)	3.31 (2); 0.21	28 (59.57%)	15 (50.00%)	0.70 (2); 0.77	24 (52.17%)	18 (60.00%)	2.60 (2); 0.30
Few	20 (25.97%)	10 (38.46%)	9 (20.00%)		11 (23.40%)	9 (30.00%)		15 (32.61%)	5 (16.67%)	
≥ Half	14 (18.19%)	3 (11.54%)	10 (22.22%)		8 (17.03%)	6 (20.00%)		7 (15.22%)	7 (23.33%)	
Emotional communicated	9 (11.69%)	2 (7.69%)	7 (15.56%)	- ^a ; 1.00	7 (14.89%)	2 (6.67%)	- ^a ; 0.46	5 (10.87%)	4 (13.33%)	- ^a ; 1.00
Physical experienced	25 (32.47%)	7 (26.92%)	16 (35.56%)	0.34 (1); 0.56	19 (40.43%)	6 (20.00%)	3.16 (1); 0.08	12 (26.09%)	13 (43.33%)	2.25 (1); 0.13
Physical asked										
Never	52 (67.53%)	16 (61.54%)	31 (68.89%)	0.83 (2); 0.66	34 (72.34%)	18 (60.00%)	1.35 (2); 0.52	31 (67.39%)	20 (66.67%)	4.19 (2); 0.13
Few	16 (20.78%)	7 (26.92%)	8 (17.78%)		8 (17.02%)	8 (26.67%)		12 (26.09%)	4 (13.33%)	
≥ Half	9 (11.69%)	3 (11.54%)	6 (13.33%)		5 (10.64%)	4 (13.33%)		3 (6.52%)	6 (20.00%)	
Physical communicated	3 (3.90%)	1 (3.85%)	2 (4.44%)	- ^a ; 1.00	3 (6.38%)	0 (0.00%)	- ^a ; 0.55	0 (0.00%)	3 (10.00%)	- ^a ; 0.22
Social experienced	13 (16.88%)	5 (19.23%)	6 (13.33%)	- ^a ; 0.75	9 (19.15%)	4 (13.33%)	- ^a ; 0.76	6 (13.04%)	7 (23.33%)	1.26 (1); 0.26
Social asked										
Never	55 (71.43%)	18 (69.23%)	32 (71.11%)	0.63; 0.80 ^a	34 (72.34%)	21 (70.00%)	0.34; 0.93 ^a	32 (69.57%)	22 (73.33%)	6.98; 0.02 ^a
Few	11 (14.29%)	5 (19.23%)	6 (13.33%)		6 (12.77%)	5 (16.67%)		10 (21.74%)	1 (3.33%)	
≥ Half	11 (14.28%)	3 (11.54%)	7 (15.56%)		7 (14.90%)	4 (13.33%)		4 (8.69%)	7 (23.33%)	
Social communicated	4 (5.19%)	2 (7.69%)	2 (4.44%)	- ^a ; 1.00	2 (4.26%)	2 (6.67%)	- ^a ; 0.53	2 (4.35%)	2 (6.67%)	- ^a ; 1.00
Medical care caregiving concern experienced	17 (22.08%)	3 (11.54%)	12 (26.67%)	2.41 (1); 0.12	11 (23.40%)	6 (20.00%)	0.08 (1); 0.78	11 (23.91%)	6 (20.00%)	0.16 (1); 0.69
Medical care caregiving concern asked										
Never	40 (51.95%)	14 (53.85%)	23 (51.11%)	0.86 (2); 0.65	25 (53.19%)	15 (50.00%)	0.36 (2); 0.86	20 (43.48%)	19 (63.33%)	7.59 (2); 0.02
Few	17 (22.08%)	6 (23.08%)	8 (17.78%)		11 (23.40%)	6 (20.00%)		15 (32.61%)	2 (6.67%)	
≥ Half	18 (23.37%)	5 (19.23%)	13 (28.89%)		10 (21.28%)	8 (26.67%)		9 (19.57%)	9 (29.99%)	
Medical care caregiving concern communicated	11 (14.29%)	3 (11.54%)	8 (17.78%)	- ^a ; 0.51	7 (14.89%)	4 (13.33%)	- ^a ; 1.00	8 (17.39%)	3 (10.00%)	- ^a ; 0.57
Other caregiving concerns experienced	17 (22.08%)	4 (15.38%)	12 (26.67%)	0.58 (1); 0.45	12 (25.53%)	5 (16.67%)	0.59 (1); 0.44	11 (23.91%)	5 (16.67%)	0.54 (1); 0.46
Other caregiving concerns asked										
Never	52 (67.53%)	16 (61.54%)	31 (68.89%)	1.14; 0.66 ^a	35 (74.47%)	17 (56.67%)	4.99 (2); 0.09	31 (67.39%)	20 (66.67%)	2.23 (2); 0.35
Few	10 (12.99%)	5 (19.23%)	5 (11.11%)		3 (6.38%)	7 (23.33%)		8 (17.39%)	2 (6.67%)	
≥ Half	14 (18.18%)	4 (15.38%)	9 (20.00%)		9 (19.16%)	5 (16.33%)		7 (15.22%)	7 (23.33%)	
Other caregiving concerns communicated	4 (5.19%)	1 (3.85%)	3 (6.67%)	- ^a ; 1.00	3 (6.38%)	1 (3.33%)	- ^a ; 1.00	2 (4.35%)	2 (6.67%)	- ^a ; 0.55

^aWhen > 20% of expected cell counts were < 5, Fisher's exact test (two-sided) was reported

Table 4 Illustrative quotes ($n=8$)

Code	Ref.	Illustrative quote
Processes	A	They come in and ask me if I needed anything or if there's anything they can get me, and they actually called me too on my phone and asked if we needed anything. (226, active treatment, late-stage gastrointestinal cancer, rural)
	B	Interviewer: During your daughter's treatment at Markey, how, if at all, has the care team checked in with you about your own needs as a caregiver? Interviewee: I, not as much. I mean, they would look, ask me if I had any questions, but as far as individually discussing it with me, that really didn't happen. (517, post-treatment, early-stage gynecologic cancer, rural)
	C	I know everything we used, we used the infusion. We were able to call them, call that number and ask questions. We could use them for when he went and wasn't feeling well, we could get extra IV fluids. I talked to a nurse practitioner there a couple times when he was really having a lot of nausea and she was going over some things to try and offered a dietician. And I don't remember a dietician coming in, but we probably said we think we're okay because she had given us different things to try. We had been offered acupuncture also, and [patient] kept saying he's going to do it, but he hadn't done it yet. But he needs to do that because of the neuropathy from the chemotherapy that he received. But he was placed on gabapentin and that helped tremendously. (219, active treatment, late-stage gastrointestinal cancer, non-rural)
	D	Interviewer: Do you remember how or if at all the care team checked in with you about what your needs might be as a caregiver? Interviewee: I do remember that her provider was very attentive and explained things very well and made sure that we were all involved and able to ask questions very good in that respect to make sure that we were, all of our concerns were validated and our questions were answered, same when she was admitted to the hospital. And the nursing staff there did a wonderful job of just making sure that everything was answered and taken care of, not just her. (524, post-treatment, early-stage gynecologic cancer, X)
	E	It's not like that he spent five minutes, he stayed there for almost a half an hour, just breaking things down and... Yeah. Yeah. He wasn't talking just to her. He was talking to both of us. (170, active treatment, late-stage unknown primary cancer, non-rural)
Internal barriers	F	The coming back for the surveillance imaging has been frustrating some days because we may be first thing over at [Building A], and then we got to go back over to [Building B], and then we're going back over to [Building C] to talk to someone. (219, active treatment, late-stage gastrointestinal cancer, non-rural)
Internal facilitators	G	Every time you go in there for chemo, they're so sweet... every single person, every single time would say, can I get you something? Do you want a pillow? Do you need a blanket? And that just really, when you're losing sleep and trying to work and help care for your family member, a cup of coffee meant a lot. And it was nice just to be asked, do you need anything? (219, active treatment, late-stage gastrointestinal cancer, non-rural)
	H	We have the ability, if I see something that's completely different from her normal activities, I can call in and they'll make the call and ask her if she's having these troubles or that it just doesn't have to be the patient. They have a good support system for the caregivers and the patients. (408, post-treatment, early-stage breast cancer, rural)
External barriers	I	We talked about it, but we're 112 miles away from you. It takes two hours to get up. It takes us two hours. Just one way. (170, active treatment, late-stage unknown primary cancer, non-rural)
	J	Well, like I said, I was[n't] always with her because I had to work. The way my FMLA runs is I only got so much time to be able to take care of her because once that started, well my work, they take your vacation the same as your FMLA. So once you run out of vacation, if they're still care for the person, there's no income. So it forces you go back to work and you're not able to provide unless you have family members to help you help. But we were fortunate enough to have enough people to care for her so I could work. (369, active treatment, early-stage breast cancer, non-rural)
External facilitators	K	At this time, WellCare has been doing well, good at helping us get rides. If something happened to my ride or daughter-in-law, my daughter-in-law, they would give us gas money or whatever. (170, active treatment, late-stage unknown primary cancer, non-rural)
Suggestions and recommendations	L	Interviewee: Yeah, I think that would be great because like I said, we don't ask for nothing. Not many people will ask for stuff Interviewer: And how do you think we could get around the barriers or challenges with asking for stuff or things like that? Interviewee: I think y'all should do follow-up calls or something just to see what kind of supports they need or if there's something you think they need to just reach out to them and see if they want it. (226, active treatment, late-stage gastrointestinal cancer, rural)
	M	I think she actually would've done really good with more of more integrative medicine services. I don't know that that was offered maybe as much as it should have acupuncture and things like that for her stress level. I mean, obviously mental health is important when you're going through something like that. She did already have a therapist before she was diagnosed, but I don't remember that really ever being asked either. So having a little bit more kind of options presented or at least checking in on that a little bit more. (524, post-treatment, early-stage gynecologic cancer, unknown)

In terms of improving survivorship services, participants recommended more personalized information delivery about treatments and services relevant to caregivers (e.g., offering videos on topics like post-surgical care) and suggested a more proactive approach to offering resources relevant to patient and caregiver needs (Table 4, L–M). Caregivers recommended increasing direct support for families, such as offering caregiver support groups and increasing tangible resource assistance like meal coupons for families helping loved ones at the hospital.

Discussion

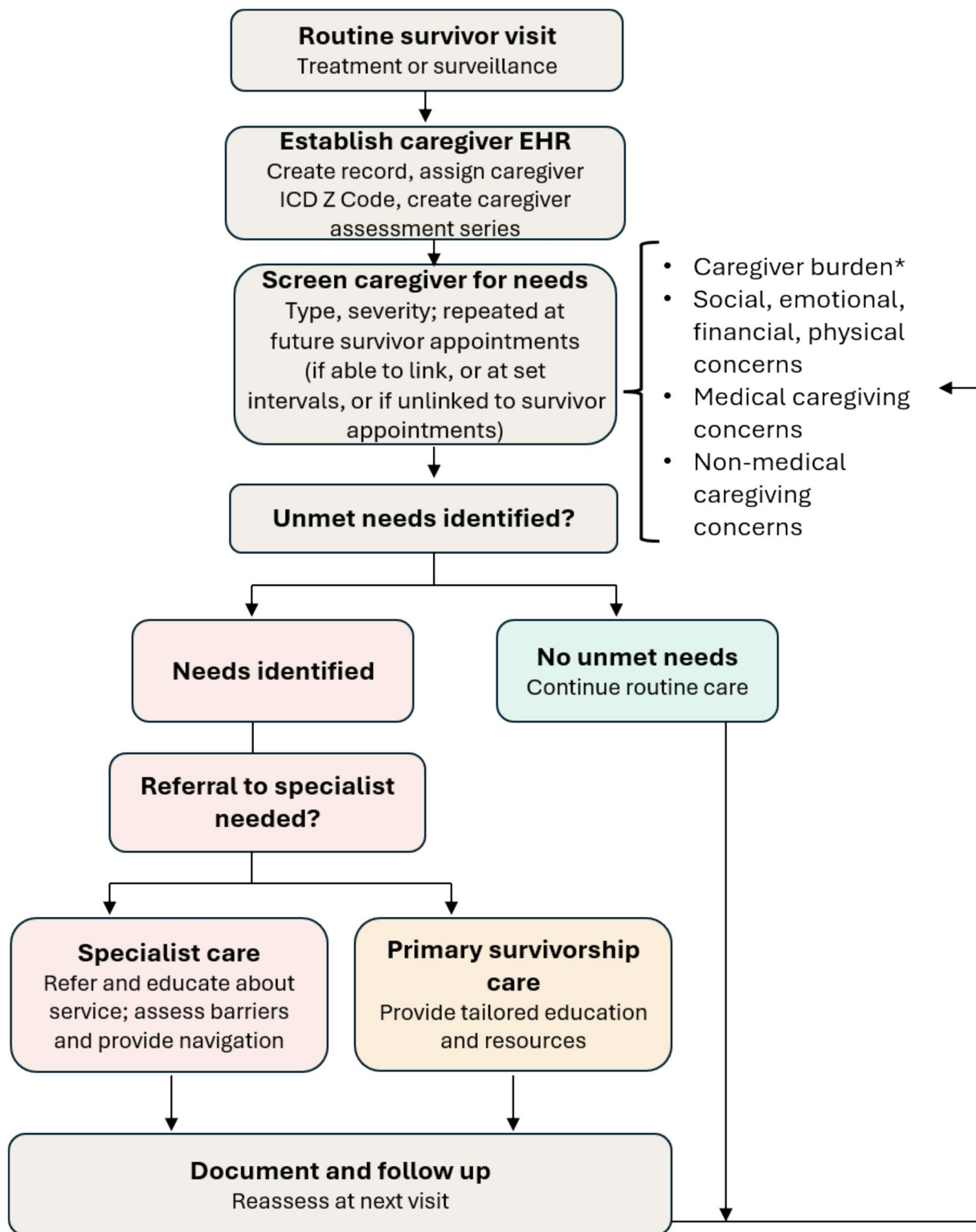
As the number of people living with a history of cancer diagnosis continues to grow [30, 31], it is paramount that the healthcare system recognizes the essential role of cancer caregivers and has procedures in place to proactively assess and manage caregiver needs. In an important step forward, the National Standards for Cancer Survivorship Care state that organizations should collect data on “caregivers’ experiences and unmet needs,” though which experiences and unmet needs should be assessed remain to be specified [14]. This study provides an example of how to assess caregiver-reported attainment of this Standard. To aid comparison with anticipated reports of survivor assessment practices included in the Standards, this study examined whether caregivers at an NCI-Designated Comprehensive Cancer Center experienced were assessed for physical, emotional, social, and caregiving-specific concerns, as well as barriers and facilitators to managing concerns. Concerns evaluated included those identified in the Standards for survivor assessment and management that are also applicable to caregivers (e.g., emotional), as well as caregiving-specific concerns (e.g., providing medical care, communicating with the medical team, managing finances, taking on additional household responsibilities) from the literature [10]. Results underscore the need for systems to consider in parallel comprehensive caregiver assessment and management practices when building practices to meet the national standards’ survivor assessment requirements.

Between 17 and 66% of caregivers reported experiencing a physical, emotional, social, or caregiving-specific concern, yet over half reported never being asked informally (i.e., in conversation) or formally (i.e., through a survey/form) by the cancer treatment team about these concerns. This finding was corroborated by the interview data and highlights a major overall gap in caregiver survivorship care, as well as suggests priorities for caregiver assessment domains. Not surprisingly, emotional concerns were the most experienced and asked about (albeit still less than 50%) concern. This finding could reflect potential downstream effects of implementation of survivor distress screening [32–34], which

are encouragingly now being applied to caregivers in some settings [35]. Physical concerns (e.g., fatigue, pain, sleep problems) were the second most common concern. The frequency of physical concerns in this sample suggests it is an area that practices should attend to for assessment, particularly for those in active treatment for whom co-occurring emotional concerns and overall burden may be greater [3, 15]—a finding further corroborated by this study’s interview data. Concerns such as pain, fatigue, and sleep disruption may be addressed through similar behavioral interventions given to survivors [36–38], as well as caregiver-tailored interventions. However, absent symptoms that meet diagnostic criteria for mood or other disorders—and until caregiver burden is recognized as a diagnostic and billing code—barriers to reimbursement for services will limit access [39], underscoring a chronic implementation problem in caregiver supportive and psychosocial care that needs to be addressed through policy.

While we expected caregivers to not report routinely being asked about their emotional, physical, and social concerns given survivor-focused cancer care, we did not anticipate that caregivers would report being infrequently asked about caregiving-specific concerns given their clear implications for survivor care. Critically, communication of concerns ranged from 4 to 14%. This may reflect, in part, a methodological artifact of survey language (i.e., “...has the cancer treatment team asked (either in conversation or a form)...”) that potentially primed caregivers to think of different members of the cancer care team (i.e., provider vs. social worker vs. nurse) who may vary with their tendency to ask [40]. Importantly, though, this result likely also underscores a stark reality in most cancer care settings: cancer care has yet to successfully implement structured processes to routinely identify, engage, and surveil caregivers, leaving caregivers to navigate survivorship care without meaningful integration within clinical workflows or timely access to tailored resources [32, 41]. This practice gap is particularly pressing in the modern therapeutic era of advanced stage cancer where caregiver responsibilities are increasing, and disease courses are lengthening [4, 6, 42, 43]. Given how infrequently caregivers reported communicating their concerns—caregiving or not—to the treatment team, it is imperative that the onus be on the health system and not the caregiver to identify caregivers’ concerns and address them satisfactorily. Indeed, without proactive assessment, caregivers’ guilt, fear of burdening the survivor, and/or worry about taking time away from a visit in voicing their own needs will be difficult to overcome [44].

The shift towards systematic, comprehensive assessment of caregiver concerns is critical to ensure broad reach of assessment, a precursor for management of concerns. While sample sizes within strata were small and only two concern domains statistically significant, a pattern emerged from



*Screen for caregiver burden to support rationale for establishing caregiver burden as a diagnostic and billing code, an important step toward increasing access to downstream caregiver services, as articulated by Applebaum and Sannes, 2025.

Fig. 2 Example assessment and referral pathway for future implementation research

survey data in this study that suggested rural caregivers may be less likely to be routinely asked (i.e., asked at least half of visits) about their concerns compared to non-rural caregivers. The difference in proportion between non-rural and rural caregivers reporting they were routinely asked about concerns ranged from ~8 to 15%, with statistically significant differences in being asked about social concerns and medical caregiving-related concerns. Previous research with cancer care administrators and clinicians suggests rural caregivers' needs are often not assessed [25], supporting the interpretation that rural caregivers may be less likely to be routinely asked about concerns in general. It is also possible that knowing a survivor and caregiver come from a rural area, cancer care teams may naturally focus conversation more on logistical and practical concerns related to barriers to care (e.g., transportation) given their high prevalence in rural populations and impact on survivor care [25, 45, 46]. Over 20% of U.S. cancer caregivers live in rural areas [10]; rural caregivers often have more unmet needs than their care recipients [47]. Systematic assessment and referral represents an opportunity to help address known barriers rural caregivers face communicating with the healthcare team [19, 48], as well as factors that may underlie constrained disclosure of concerns (e.g., stigma about seeking help, cultural value of self-reliance [49, 50]) among rural residents.

Qualitative data from this study suggest several important considerations for extending standards to caregivers and planning for their implementation. Beyond establishing routine assessment of caregiver concerns as part of survivorship care, standardized referral pathways to on-site and local services need to be developed, along with education about recommended services and navigation and support to reduce barriers to utilization (Fig. 2). Several other cancer care delivery initiatives, including distress screening and tobacco screening [33, 51–53], may offer a helpful framework for procedures that could be adapted. Leveraging implementation science frameworks such as Fit-to-Context Framework for Designing for Dissemination and Sustainability [54] and co-creation practices [55] where multilevel partners are engaged could help ensure procedures are feasible in specific settings, have maximal reach and effectiveness, and can be sustained as the field moves to reframe the survivor-caregiver as the unit of care in cancer care and align payment models [56]. The National Standards for Cancer Survivorship Care offer an important step in this mandate for change.

The caregiver sample in this study was predominantly White, educated, and relatively affluent [57–59], which may have influenced the types of challenges caregivers experienced, their ability to communicate concerns, and access to supportive resources. Consequently, these findings may not generalize to caregivers with fewer socioeconomic resources, who may face additional barriers [59]. Other limitations include the single-site design and self-reported measures,

which are subject to recall or social desirability bias. This study relied on caregiver reports of whether concerns were discussed with the care team and did not evaluate formal health system processes for assessment and management of caregiver needs. Future research should continue to examine how caregiver assessment is operationalized within clinical workflows and how identified needs are linked to supportive services [32, 60]. While the mixed-methods design provides depth, the qualitative sample was relatively small, thematic saturation was not formally assessed, and additional interviews might have revealed further nuance. Additionally, while this study evaluates caregiver experiences relevant to the National Standards for Cancer Survivorship Care, the standards do not define specific caregiver assessment procedures; therefore, our findings should be interpreted as informing potential approaches for caregiver assessment rather than evaluating implementation fidelity to a prescribed caregiver assessment model. Indeed, we intentionally paralleled the survivor-recommended domains based on the premise that these needs are equally essential to caregiver well-being.

One of the first studies to assess alignment of the National Standards for Cancer Survivorship Care, this study offers meaningful insights into the current state of survivorship care from the caregiver perspective. Future research should examine multi-site and diverse samples, identify subgroup differences (e.g., rural vs. non-rural, lower vs. higher socioeconomic status), and test implementation strategies that operationalize caregiver inclusion within survivorship care workflows. Policy and practice initiatives should prioritize caregiver assessment and support as core components of survivorship care. Incorporating National Survivorship Standards [14] into clinical practice guidelines for identifying caregivers and assessing their needs could strengthen care delivery.

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Data availability Survey data are available upon reasonable request to the corresponding author. The qualitative data underlying this article cannot be shared publicly due to the desire to protect the privacy of individuals who participated in the study. Deidentified interview transcripts will be shared upon reasonable request to the corresponding author.

Declarations

Ethics approval This study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the University of Kentucky Institutional Review Board (IRB #98262).

Consent to participate Informed consent was obtained from all individual participants included in the study.

Competing interests The authors declare no competing interests.

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